| | | | |
|---|---|---|---|
| 7. | *Ephedra gerardiana* | Ephedrine | For the treatment of asthma, bronchititis |
| 8. | *Pinus roxburghii* | Oleoresin | Used to make soap, varnishes and printing ink |
| 9. | *Pinus roxburghii, Picea smithiana* | Wood pulp | Used to make papers |
| 10. | *Cedrus deodara* | wood | Used to make doors, boats and railway sleepers |
| 11. | *Cedrus atlantica* | oil | Used in perfumery |
| 12 | *Thuja, Cupressus, Araucaria, and Cryptomeria* | whole plant | Ornamental plants/Floral Decoration |

### 2.6.5 Cycas

Class – Cycadopsida
Order – Cycadales
Family- Cycadaceae
Genus - *Cycas*

It is widely distributed in tropical and sub tropical region of eastern hemisphere of the world. *Cycas revoluta*, *Cycas beddomei*, *Cycas circinalis, Cycas rumphii* are some of the common species. The plant body is sporophyte and resemble a small palm. The growth is very slow. It is evergreen and xerophytic in nature.

**Sporophyte**

The sporophyte is differentiated into root, stem and leaves. The stem is columnar bearing a crown of spirally arranged pinnately compound leaves (Figure 2.23).

**External features**

**Root**

Two types of roots are found in *Cycas*. They are the tap root and coralloid root.



**Figure 2.23:** *Cycas* Habit

The primary root persists and forms the tap root.

Some of the lateral roots give rise to branches which grow vertically upward below the ground level. They branch repeatedly to form dichotomously branched coral- like roots called coralloid roots. The cortical region of the coralloid root contains the Blue green alga – *Anabaena* sp. which helps in nitrogen fixation (Figure 2.24).

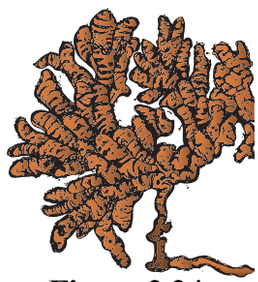


**Figure 2.24:** Coralloid root

**Stem**

The stem is columnar, unbranched and woody. It is covered with persistent woody leaf bases. The stem also bears adventitious buds at the base.

**Leaves**

*Cycas* has two types of leaves

(i) Foliage or assimilatory leaves

(ii) Scale leaves

(i) Foliage or assimilatory leaves

Foliage leaves are large, pinnately compound and form a crown at the top of the stem. Each leaf has 80-100 pairs of sessile leaflets. The apex is acute or spiny. The leaflet has a single midvein. Lateral veins are absent. Circinate vernation is present and young leaves are covered with **ramenta**.

(ii) Scale leaves

Scale leaves are brown, small, triangular and persistent which are protective in function. They are covered with ramenta.

**Internal structure**

**T.S. of Root**

The internal organization of the primary root reveals the following parts. 1. Epiblema,



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